Assignment – 1

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1. What is Web Development?

Web development is the process of designing, creating, and maintaining websites and web applications that run on the internet or private networks. It involves both the visual presentation of a website and the underlying logic that controls how the website functions.

Web development includes activities such as:

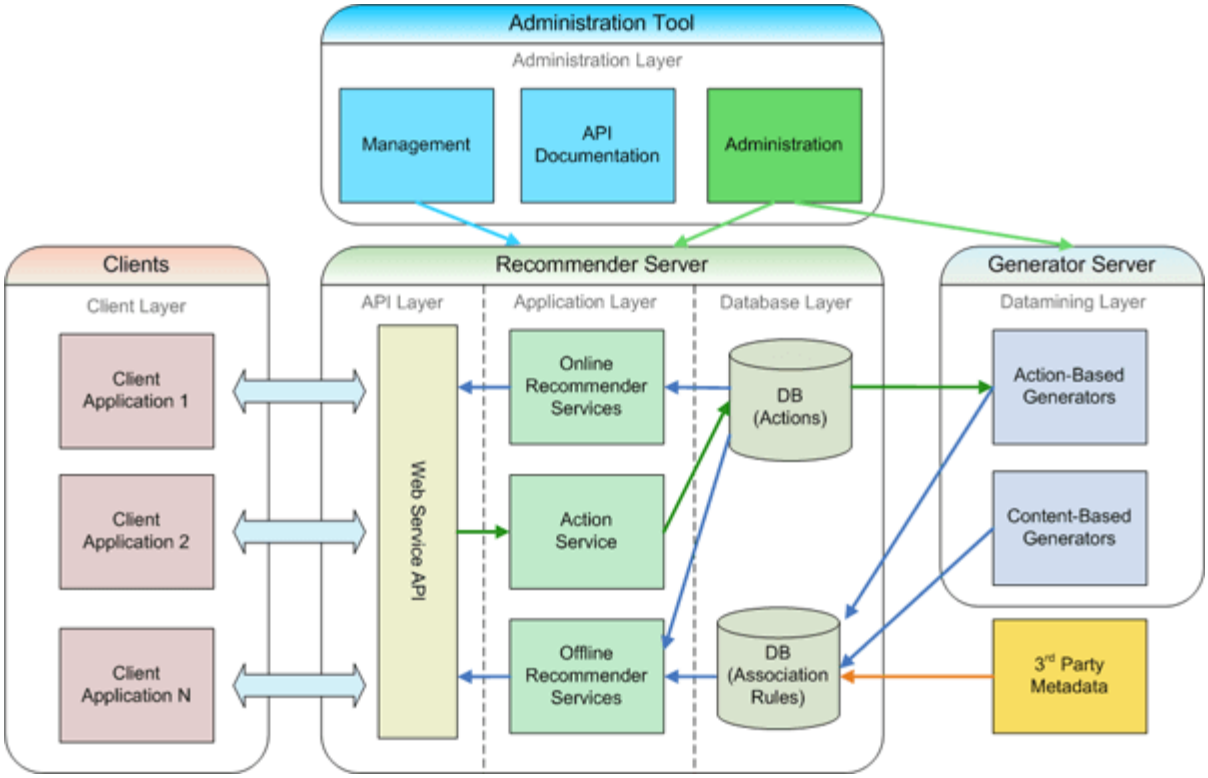
- Designing layouts and user interfaces
- Writing client-side and server-side code
- Managing databases and data flow
- Deploying applications on web servers

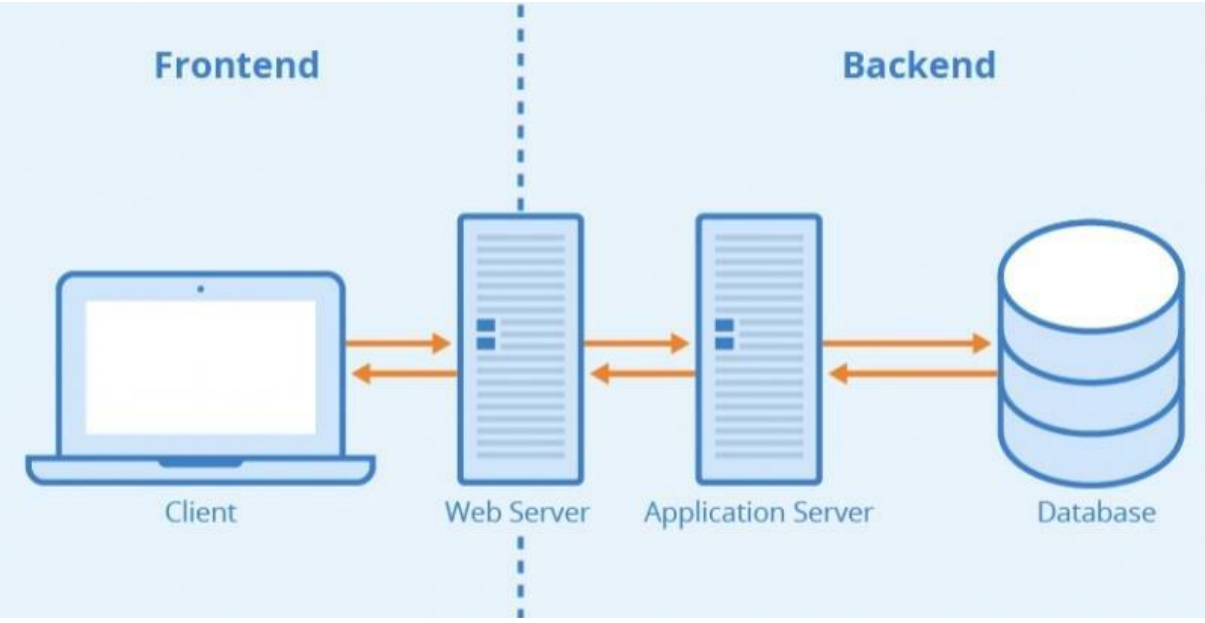
A complete web application works through the coordination of the frontend, backend, and database. Web development is widely used in applications such as online banking, e-commerce websites, social media platforms, and cloud-based systems.

Web development is broadly classified into:

- Frontend development
- Backend development
- Full stack development

Web Development Architecture Diagram





Explanation:
The frontend handles user interaction, the backend processes requests and business logic, and the database stores application data. All components work together to deliver a functional web application.

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2. Difference between Frontend and Backend (with examples)

Frontend and backend development represent two different layers of a web application. Both are essential and depend on each other to ensure smooth operation of a website.

Frontend Development

Frontend refers to the user-facing part of a website. It focuses on design, layout, and user interaction. Everything that users see on their screen, such as buttons, forms, and menus, is part of the frontend.

Backend Development

Backend refers to the server-side part of a website. It handles data processing, business logic, authentication, and communication with the database.

Frontend	Backend
Visible to users	Not visible to users
Runs in browser	Runs on server
Focuses on UI/UX	Focuses on logic and data
Uses HTML, CSS, JavaScript	Uses Node.js, Java, Python
Example: Login page design	Example: Login validation

Example:
When a user submits a login form, the frontend sends the data to the backend. The backend checks the credentials in the database and returns a success or failure response.

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3. Client–Server Communication

Client–server communication is a network model in which a client (such as a web browser) requests services or data from a server, and the server responds after processing the request. This communication is mainly done using HTTP or HTTPS protocols.

In this model:

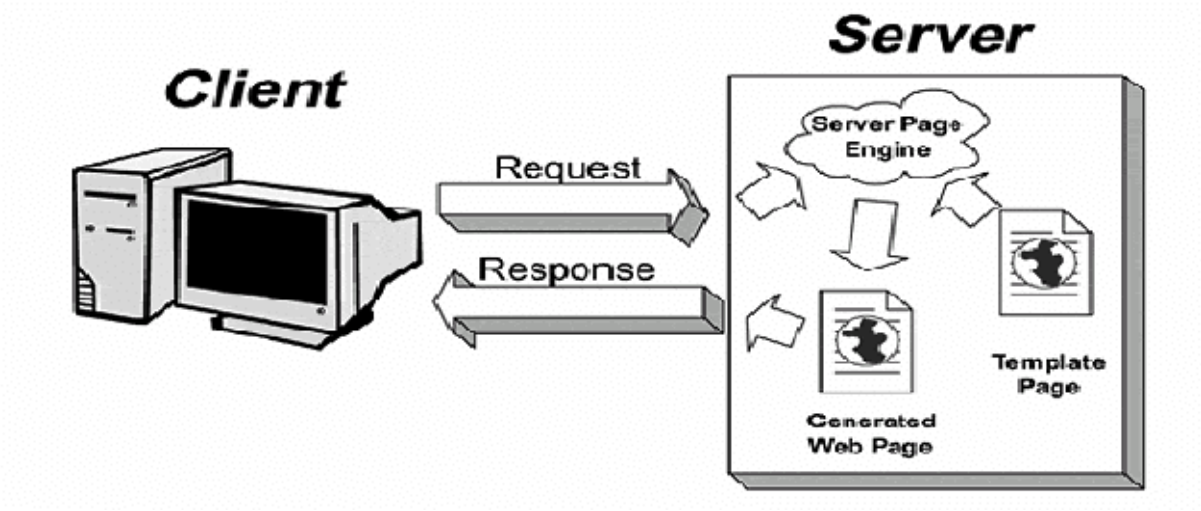
- The client initiates the request
- The server listens and responds
- Data flows in the form of requests and responses

Communication Process

1. Client sends an HTTP request
2. Server receives and processes the request
3. Server accesses the database if required

4. Server sends an HTTP response back to the client

Client–Server Communication Diagram



This model allows multiple clients to communicate with a single server efficiently.

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4. What is MEAN Stack?

MEAN stack is a full-stack JavaScript framework used to build dynamic and scalable web applications. It allows developers to use JavaScript across both frontend and backend, making development faster and more consistent.

MEAN stands for:

- **MongoDB** – A NoSQL database that stores data in JSON-like documents
- **Express.js** – A backend web framework for handling routes and APIs
- **Angular** – A frontend framework for building dynamic user interfaces
- **Node.js** – A runtime environment that executes JavaScript on the server

Advantages of MEAN Stack

- Single programming language (JavaScript)
- High performance and scalability
- Easy maintenance
- Suitable for real-time applications

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5. Install Angular CLI using npm

Angular CLI (Command Line Interface) is a tool that helps developers create, manage, and build Angular applications efficiently. It reduces manual work by providing predefined commands.

Installation Command

npm install -g @angular/cli

Explanation

- npm is Node Package Manager
- -g installs the package globally
- Angular CLI helps in creating components, services, and modules

Verify Installation

ng version

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6. MEAN Architecture Workflow

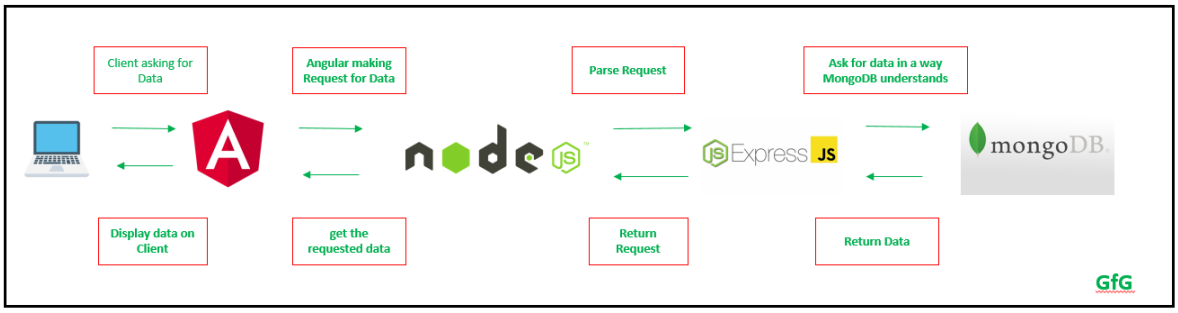
The MEAN stack follows a layered architecture where each component has a specific role and communicates with other components through HTTP requests.

Workflow Description

1. User interacts with the Angular frontend
2. Angular sends HTTP request to Express server
3. Express handles routing and logic using Node.js
4. Node.js communicates with MongoDB to fetch or store data
5. Server sends response back to Angular frontend

This architecture ensures separation of concerns and better scalability.

MEAN Stack Architecture Diagram



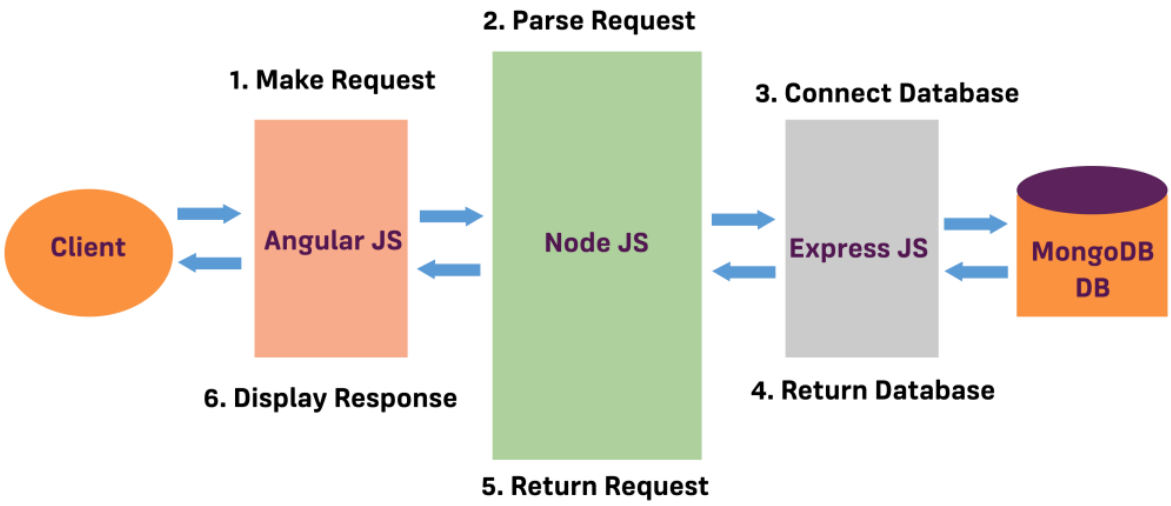


Fig: Mean Stack App Architecture